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MODEL REPORT

CYP direct-inhibition challenge

Report detailing pIC50 regression models for CYP1A2, CYP2C9, CYP2D6, and CYP3A4 isoforms

Models developed as part of the OpenADMET CYP Blind Challenge

Team name: EleanorRigby

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1. Summary

This document describes the methodology used to develop four independent CYP isoform models. Each model is a weighted blend of two learners over a shared molecular feature block. Predictions are subjected to isoform-specific post-hoc supervised transformations.

Across the challenge, we evaluated 8 different learning algorithms and 19 molecular representations. This breadth of exploration would not have been feasible without agentic AI, which we used extensively to develop, execute and evaluate our code throughout the challenge.

2. Data

Training data used to build the models is comprised of the challenge release plus public domain. No proprietary CYP data was used.

Table 1 – Training data sources

source	file / origin	content	how it is used
Dose-response labels	cyp-challenge-TRAIN_inhibition.csv	4,905 molecules × 4 isoforms; per-isoform n = 1,412 (1A2), 1,285 (2C9), 1,493 (2D6), 2,335 (3A4)	regression modeling
Single-concentration screen	cyp-challenge-single-concentration-TRAIN.csv	17,504 rows, 4,376 molecules across 4 CYPs	(a) isotonic source for pseudo labelling; (b) a multi-task head

ChEMBL, target-first pull	155,803 activities / 39,695 molecules / 14,988 assays → 15 curated heads, 72,767 labels, 39,237 molecules	PubChem qHTS potency, NCATS panel pAC50, literature IC50/Ki/AC50	multi-task heads

There are four blinded molecules in public domain. The target-first ChEMBL pull in Table 1 recovers four of the 750 blinded molecules. The data for these are listed below in Table 2. All seven rows were masked from our training set before heads were built. The coverage of these four molecules yields values on CYP1A2 and CYP2C9, upper bounds on CYP2C9 and CYP2D6, and nothing for CYP3A4.

Table 2 – The four blinded molecules in ChEMBL and associated data.

blinded molecule	ChEMBL ID	isoform	assay	reported activity
OCNT-0496981	CHEMBL1352528	CYP1A2	`CHEMBL1741322`	AC50 1,584.89 nM = pAC50 **5.80**
OCNT-0496981	CHEMBL1352528	CYP2C9	`CHEMBL1741325`	AC50 1,584.89 nM = pAC50 **5.80**
OCNT-0496981	CHEMBL1352528	CYP2D6	`CHEMBL1741321`	AC50 11,220.18 nM = pAC50 **4.95**
OCNT-0496981	CHEMBL1352528	CYP3A4	`CHEMBL1741324`	row contains **no value**
OCNT-2535297	CHEMBL3960951	CYP2C9	`CHEMBL3867191`	IC50 **> 10,000 nM**
OCNT-2535303	CHEMBL3930818	CYP2C9	`CHEMBL3867191`	IC50 **> 10,000 nM**
OCNT-2535788	CHEMBL3893879	CYP2C9	`CHEMBL3867191`	IC50 **> 10,000 nM**

Curation of ChEMBL heads

- All seven rows in Table 2 masked by canonical SMILES before auxiliary heads built to remove each molecule.
- qHTS potency head restricted to `activity_comment == 'Active'`; CYP3A4 activator assay excluded entirely.
- The NCATS panel carries 36,866 value-free rows and 13,550 explicitly inconclusive ones; only unambiguous Active rows with relation `=` added.
- ChEMBL rows are merged onto challenge-set molecules by canonical SMILES, so fold hold-out removes molecules whole.

Pseudo-labelled actives / inactives. An isotonic regression fit to the relationship between –log2fc against measured pIC50 in the challenge data set. This fit was then used to predict pseudo-pIC50 values for screening molecules lacking dose-response data. Molecules with pseudo-pIC50 values below 4.3 were classified as inferred inactives.

All four isoform models benefited from inclusion of the inferred inactives. Two models, CYP1A2 and CYP3A4, also benefited from inclusion of the pseudo-labeled actives. For CYP3A4, pseudo-labels were corrected time-dependent inhibition (TDI), as described in §5.3.

The incorporation of this pseudo-labeled data provided the single largest improvement in overall model performance.

Table 3 – Pseudo-labelled data per isoform.

isoform	pseudo rows used	pseudo max	pseudo ≥ 4.3	isotonic fitted on
CYP1A2	2,963	5.444	639	pooled, all isoforms
CYP2C9	1,426	4.283	0	own isoform
CYP2D6	2,882	4.246	0	pooled
CYP3A4	2,570	5.719	1,555	own isoform

3. Shared representation – the 406-column feature block

Identical columns for all four isoforms; PCA fitted on train unsupervised, no labels.

Table 4 – The 406-column feature block.

block	width	content
RDKit descriptors	202	the full RDKit descriptor list; the 512 Morgan-fingerprint excluded
Basicity pharmacophore	12	SMARTS counts for basic aliphatic N, amidine, guanidine, pyridine N, quaternary N, aniline N, amide N; plus has-basic, total basic N, min and mean topological distance from a basic N to any aromatic atom, and basic-N fraction of all N
Pretrained encoder, PCA-192	192	surrogate_adme message-passing encoder from OpenADMET; frozen, mean-aggregated, PCA to 192

4. Methods used to generate predictions

All weights fixed and not fitted on a held-out splits.

Table 5 – The four isoform models

	CYP1A2	CYP2C9	CYP2D6	CYP3A4
Labelled rows	**1,412**	**1,285**	**1,493**	**2,335**
Augmentation	**2,963**	**1,426**	**2,882**	**2,570 TDI-corrected**
TabICL learner	TabICL, 406 cols	same	same	same
Hedge weight	**0.80**	**0.75**	**0.85**	none
aux-GNN learner	35 heads, CheMeleon backbone, scored head trained on the augmented target	35 heads, surrogate_adme backbone	none	surrogate_adme, TDI-classified molecules dropped, CYP2D6 data masked (14 live heads of 35)
Blend	**0.75 TabICL + 0.25 aux-GNN**	**0.65 TabICL + 0.35 aux-GNN**	**TabICL only**	**0.65 TabICL + 0.35 aux-GNN**

As ST-RAE was used for leaderboard ranking it was necessary to transform model outputs. All transforms were performed in supervised manner by inspection of model output.

Table 6 – Post-hoc transforms

	CYP1A2	CYP2C9	CYP2D6	CYP3A4
Shaping / placement	monotone non-affine recalibration	monotone non-affine recalibration	monotone recalibration + compression toward mean	monotone recalibration + compression

5. Implementation details

5.1 TabICL and the hedge

TabICLRegressor on the 406 columns in Table 4. Two fits per isoform: one on the **inactives** population (measured + pseudo rows), and one on the **actives** population (measured + pseudo rows). Predictions on the blinded 750 are averaged at the hedge weight, w, given in Table 5:

```
TabICL = w · TabICL(measured + pseudo) + (1 − w) · TabICL(measured + pseudo)
```

The above hedge was implemented because the two populations disagree about the right model.

5.2 The auxiliary multi-task GNN

chemprop 2.3.1 `MPNN = BondMessagePassing` (hyper-parameters taken verbatim from the pretrained checkpoint, `strict=True` load) + `MeanAggregation` + `RegressionFFN` with 35 task heads and NaN-aware per-task standardisation. 5-fold `KFold(shuffle=True, random_state=0)` over the target rows, 40 epochs, early stopping patience 12, Noam schedule, `batch_norm=False`. Auxiliary task weight fraction 0.1 (0.3 measured worse). Trained on the union of the challenge molecules and the ChEMBL pull: **35 heads, 111,473 labels, 45,384 molecules** (6,147 challenge + 39,237 ChEMBL-only + 251 merged).

Head families: four scored `pIC50_{ISO}` heads; `log2fc_{ISO}` (the screen); `emax_di_{ISO}` and `emax_tdi_{ISO}`; `chembl_old_{ISO}`; and the four ChEMBL families `qhts_pot`, `ncats_ac50`, `lit_pot`, `screened`.

Two different configurations are used:

- **35 live heads** (CYP1A2 and CYP2C9 slots).
- **14 live heads of 35** (CYP3A4 slot): 21 masked, comprising a 16-head family removal (`lit_pot`, `screened`, `emax_tdi`, `chembl_old`) plus all five CYP2D6 heads. Also drops 1,238 "TDI-suspect" molecules — those carrying a TDI-condition Emax with no direct-condition Emax and no scored pIC50 — leaving 44,146 molecules / 110,151 labels.

5.3 CYP3A4's TDI corrections

CYP3A4 is the only isoform where we use a TDI correction to predictions. This is mainly due to the fact that 3A4 is only isoform where we believed the output of a TDI classification model (see classification track for 3A4 model).

1. A 3A4 TDI classifier (see classification track) supplies a per-molecule TDI probability.
2. For probable TDIs, correction applied by delta-pIC50 regression model (see classification track)

5.4 Environment

Python 3.12.13, numpy 2.4.6, pandas 3.0.5, scikit-learn 1.9.0, RDKit 2026.03.5, tabicl 2.1.1, chemprop 2.3.1, lightning 2.6.5, torch 2.13.0 (Apple MPS), lightgbm 4.7.0, atom-jepa 0.3.0.